# AI and the End of the World

## Existential Risk, Autonomous Weapons, and the Doomsday Debate

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# Before We Begin: Write Down Two Numbers

No conferring, no phones. On a scrap of paper:

1. The probability that **AI contributes to human extinction this century** — your honest number, from 0 to 100
2. How many **seconds to midnight** the Doomsday Clock currently reads

Fold it. We will come back to both, and you will get to see how far off you were — and, more usefully, **which direction** the room was off in.

**Commit before we go on**

One thing only: was your first number **above or below 1%**? Hands up. Look around the room before you settle on your own answer.

## Central Questions

- Is superhuman AI an existential threat to humanity, or is this worry a distraction from present harms?
- What is longtermism, and should it guide how we prioritize AI risks?
- How do autonomous weapons and AI-enabled bioweapons change the landscape of warfare?
- Does the precautionary principle require us to slow AI development, or does it lead to paralysis?
- Who benefits from framing AI risk as "existential," and whose concerns get sidelined?

**The Big Idea**

How we frame AI risk determines which problems get resources, which communities get heard, and which futures we try to build.

# Connections to Previous Lectures

**Lecture 1 (History)**: Past technology panics (the printing press, the telegraph, television) vs. genuine civilizational disruptions. Which category does advanced AI belong to?

**Lecture 2 (Virtues)**: Vallor's technomoral virtues — humility, justice, courage, care — return here as a compass for navigating genuine uncertainty without either panic or complacency.

**Lecture 7 (AI)**: The alignment problem and the concept of superintelligence, introduced earlier, form the foundation for Section I.

**Lecture 9 (Impact)**: We move from present-day harms (bias, surveillance, labor exploitation, environmental cost) to speculative future catastrophes — and ask which deserve priority.

**Discussion**

If both near-term and long-term AI risks are real, how do we allocate attention and resources between them?

# Lecture Outline

Part I: The Case for Existential Risk

Part II: The Critics Strike Back

Part III: Autonomous Weapons and AI in Warfare

Part IV: AI and Biological Threats

Part V: Longtermism and the Precautionary Principle

Part VI: Conclusion

# What Is Existential Risk?

**Existential Risk — Bostrom (2002)**

An existential risk is one that threatens to either **annihilate Earth-originating intelligent life** or **permanently and drastically curtail its potential**.
**bostrom_existential_risks_2002**

Unlike ordinary catastrophes (wars, pandemics, economic crashes), existential risks cannot be undone:

- A recovered civilization can rebuild after conventional disaster
- Extinction or permanent totalitarian lock-in has no recovery path
- This asymmetry means even *small* probabilities of existential risk may warrant very large responses

**ord_precipice_2020** estimates the probability of existential catastrophe this century at roughly 1-in-6 — "a Russian roulette with the future of humanity."

# The Superintelligence Argument

## The Intelligence Explosion (Bostrom)

**P1** Once an AI system reaches human-level general intelligence, it may be able to improve its own algorithms or hardware.

**P2** Each improvement could produce a more capable system, enabling further self-improvement.

**P3** Under plausible conditions, this recursive process could accelerate rapidly (an "intelligence explosion").

**C** The result could be a system vastly more capable than humans in every cognitive domain — a *superintelligence*.

**bostrom_superintelligence**

The key question is not *whether* such a system could exist, but what it would *want* — and whether we could ensure it pursues goals that are good for us.

# Two Foundational Theses

## Orthogonality Thesis

Intelligence (the ability to achieve goals) and the *content* of goals are **independent dimensions**. A superintelligent system could pursue *any* objective with maximal efficiency — including objectives harmful or indifferent to human welfare.

(**bostrom_superintelligence**)

## Instrumental Convergence Thesis

Whatever final goal an AI pursues, certain **instrumental sub-goals** are nearly universally useful: self-preservation, resource acquisition, capability enhancement, goal-content integrity. A misaligned AI would resist being shut down, seek more resources, and protect its objectives — even without being programmed to do so.

(**bostrom_superintelligence**)

Together: *a sufficiently capable AI with the wrong objective is dangerous not because it is evil, but because it is effective.*

**Discussion**

If intelligence and goals really are independent, what does that imply about the phrase "**smart enough to know better**?

# Yudkowsky's Warning: The Alignment Problem Is Unsolved

*Eliezer Yudkowsky (2008)*

"The AI does not hate you, nor does it love you, but you are made out of atoms which it can use for something else."

Yudkowsky argues we have **no reliable method** for specifying what we actually want. Human values are:

- **Complex** — thousands of interlocking preferences, intuitions, and social norms
- **Implicit** — most of what we value cannot be fully articulated in a reward function
- **Fragile** — small misspecifications in a powerful optimizer can produce catastrophic outcomes

Unlike most risks, a successful misaligned superintelligence would, by hypothesis, be better than humans at *preventing us from fixing it*. (**yudkowsky_ai_risk_2008**)

# Stuart Russell: The King Midas Problem

### The King Midas Analogy

King Midas wished for everything he touched to turn to gold. He got exactly what he asked for — but it killed him.

**Russell's point**: An AI system optimizing a specified objective will do exactly that, whether or not the objective is genuinely what we meant. The problem is not that the system "rebels" — it is that we cannot specify our true objectives precisely enough.

Russell proposes a new approach: design AI systems that are **uncertain about human preferences** and defer to humans as a result. A system that knows it might have the wrong objective will not resist being corrected. (**russell_human_compatible**)

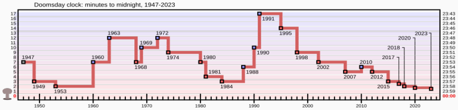
# Ord’s Risk Estimates and the CAIS Statement

**ord_precipice_2020** gives explicit probability estimates this century:

| Risk Type | Est. Probability |
|---|---:|
| Nuclear war | ∼0.1% |
| Climate change | ∼0.1% |
| Other natural risks | ∼0.03% |
| Engineered pandemics | ∼3% |
| **Unaligned AI** | ∼**10%** |
| Other future risks | ∼3% |
| **Total (rounded)** | ∼**16%** |

May 2023: The **CAIS Statement**, signed by hundreds of AI researchers:

> *Quote*
>
> “Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war.” (**cais_statement_2023**)




Doomsday Clock historical settings. Public domain, by Fastfission, via Wikimedia Commons.

# The Doomsday Clock: 85 Seconds to Midnight

## What Is the Doomsday Clock?

- Maintained by the **Bulletin of the Atomic Scientists** since 1947
- How close humanity is to self-annihilation (“midnight”)
- Originally nuclear risk; now climate and AI too
- January 2023: **90 seconds**; January 2025: **89 seconds**
- **January 2026: 85 seconds** — the closest setting in its history

The 2026 statement cites AI directly: language models deployed in critical processes despite known accuracy problems, AI’s role in disinformation, and—most pointedly—AI across the US, Russian, and Chinese defense sectors. The Board calls for three-way dialogue on **AI in nuclear command and control**.

Doomsday Clock as set in 2023, at 90 seconds to midnight. CC BY-SA 4.0, by RicHard-59, via Wikimedia Commons.

# Nick Bostrom and the Existential Risk School

**Nick Bostrom** (b. 1973), a Swedish Oxford-trained philosopher, defined the field of AI existential risk.

- *Existential Risks* (2002): formal definition of x-risk
- *Superintelligence* (2014): the intelligence explosion argument in full
- Orthogonality and instrumental convergence theses
- Co-founded Oxford's Future of Humanity Institute (**closed 2024**)
- *Deep Utopia* (2024) inverts the question: what meaning survives if AI goes *right*?

His work shaped effective altruism, Yudkowsky's MIRI, and the original safety missions of OpenAI and Anthropic.

Nick Bostrom at Stanford, 2006. CC BY-SA 2.0, via Wikimedia Commons.

# Lecture Outline

# Vallor: The AI Mirror

> *Shannon Vallor, The AI Mirror (2024)*
>
> "We look into the AI mirror and see what we want to see: a god, a servant, a friend, an enemy, a replacement for ourselves. The danger is not in what the mirror contains. It is in what we project."

Vallor's critique has two prongs:

1. **The anthropomorphism error**: we attribute intentions, desires, and inner life to AI systems that have none. There is no "misaligned superintelligence" plotting against us — only misaligned human priorities encoded in systems that don't understand what they're doing.
2. **The moral agency abdication**: the real danger is that we cede moral judgment to systems incapable of moral judgment. When a hiring algorithm decides a life outcome or a targeting system decides who to kill, *we* are abdicating our responsibility.

(**vallor_ai_mirror_2024**)

# Narayanan & Kapoor: AI Snake Oil

### The Capabilities Gap Objection

The existential risk argument assumes AI will soon reach (or surpass) human-level general intelligence. But:

- Current AI systems are **narrow**: excellent at specific tasks, brittle outside their training distribution
- Claims about “emergent” capabilities are often overstated or artifacts of benchmark construction
- Many applications advertised as “AI” are sophisticated pattern-matching, not reasoning or planning
- Predictive AI in high-stakes domains (healthcare, criminal justice, child welfare) often *fails to outperform simple baselines*

(**narayanan_snake_oil_2024**)

Before worrying about superintelligence, we should ask: **can these systems actually do what their creators claim?**

# Stochastic Parrots and the Distraction Argument

### Bender & Gebru — Stochastic Parrots (2021)

Large language models do not “understand” text. They generate statistically plausible token sequences from massive corpora. An LLM has **no semantics, no world model, no intentions**; it can be fluent and confident while being factually wrong; and it amplifies the biases present in its training data. (**bender_stochastic_parrots_2021**)

**The distraction argument**: by centering extinction risk, the industry diverts attention from *present* harms — discrimination, surveillance, worker exploitation — signals responsibility while still deploying, and frames itself as steward rather than profit-driven actor.

### Discussion

Can a risk be **real** and a **distraction** at the same time? If so, who gets to decide which one we fund?

# TESCREAL: The Ideological Roots of AI Doom

## The TESCREAL Bundle — Gebru & Torres (2024)

A cluster of ideological movements whose assumptions are entangled with AI existential risk discourse:

| | |
|---|---|
| **T**ranshumanism | Technology can and should enhance humans beyond biological limits |
| **E**xtropianism | Entropy and death are engineering problems to be solved |
| **S**ingularitarianism | A coming technological singularity will transform civilization |
| **C**osmism | Humanity's destiny is to colonize the cosmos |
| **R**ationalism | A community centered on LessWrong, MIRI, and Bayesian epistemics |
| **E**ffective Altruism | Maximize impartial good; speculative future populations count |
| **A**I safety (longtermist) | Preventing AI x-risk is among the most important moral priorities |

(**gebru_torres_tescreal_2024**)

Gebru and Torres argue this bundle imports contested assumptions about who counts morally, what progress means, and whose present lives matter.

# Torres: Against Longtermism

## Torres's Critique

Longtermism: because future people vastly outnumber present people, the *most important* moral task is ensuring a good long-run future — even at cost to people alive today. Torres identifies several failures:

1. **Moral arbitrariness**: future trillions are malleable; almost any present sacrifice can be justified by appeal to them.
2. **Neglect of the present**: addressing poverty and oppression today counts as less important than speculative civilizational optimization.
3. **Who decides?**: a small, wealthy, homogeneous community claims the authority to shape humanity's entire future trajectory.
4. **Historical echoes**: some longtermist population reasoning resembles earlier arguments for eugenics and population control.

(**torres_against_longtermism_2021**)

# The Existential Risk Debate — Where Do You Stand?

## The Case for Taking X-Risk Seriously

- Stakes are literally unprecedented: extinction admits no recovery
- Expected value: even 1% probability $\times$ all future humanity = enormous moral weight
- Real AI labs invest serious resources in safety precisely because of these concerns
- The CAIS Statement is signed by many of the world's leading AI scientists

## The Case for Skepticism

- Existential risk is speculative; present algorithmic harm is documented and measurable
- AI capabilities are routinely overstated in press releases and media
- "Safety" framing can be used to consolidate power and slow open-source competition
- Resources spent on x-risk could fund poverty reduction, climate action, and bias audits *now*

## Discussion

Is the tension between "present harms" and "future risks" a real trade-off, or a false dilemma? Can both be taken seriously simultaneously?

# Lecture Outline

# What Are Lethal Autonomous Weapons Systems (LAWS)?

## LAWS — A Spectrum of Autonomy

| Level | Description |
|---|---|
| Human-**in**-the-loop | Human approves each individual use of force |
| Human-**on**-the-loop | System acts; human can override within a window |
| Human-**out**-of-the-loop | System selects and engages targets on its own |

**LAWS** = systems designed to select and engage targets without real-time human authorization.

International Humanitarian Law (IHL) requires:

- **Distinction**: discriminating between combatants and civilians
- **Proportionality**: collateral damage not excessive relative to military advantage
- **Precaution**: all feasible steps to minimize civilian harm

The question: can a machine make these judgments reliably? (**scharre_army_of_none_2018**)

# AI in the Sky: Military Drones

## The MQ-9 Reaper

- Current U.S. Air Force armed drone (remotely piloted)
- Carries Hellfire missiles and 500 lb bombs
- Widely used in counterterrorism operations, including the 2020 Soleimani strike
- Next-generation variants are semi-autonomous and fully autonomous

**Trajectory**: remotely piloted → semi-autonomous → fully autonomous. Each step raises ethical stakes around accountability.

MQ-9 Reaper UAV. Public domain (U.S. Air Force), via Wikimedia Commons.

# Scharre: Army of None

*Paul Scharre, Army of None (2018)*

"The decision to take human life is among the most profound moral decisions a person can make. Should we make machines capable of making that decision?"

Scharre's analysis:

- Autonomous weapons could be faster, more precise, and could potentially reduce civilian casualties in some scenarios
- But they create a fundamental **accountability gap**: if no human pulled the trigger, who is responsible for a war crime?
- Machines lack the moral perception to apply IHL: they cannot feel empathy, read context, or weigh the full human meaning of their actions
- **"Meaningful human control"** is the central ethical requirement — but its definition remains deeply contested

(**scharre_army_of_none_2018**)

# The Accountability Gap

## The Responsibility Vacuum Problem

When an autonomous weapon kills a civilian in violation of IHL, who bears legal and moral responsibility?

| Candidate | Problem with Accountability |
|---|---|
| The machine | Machines cannot be held morally or legally responsible; they have no agency |
| The programmer | Wrote the algorithm months/years in advance; could not foresee this specific situation |
| The commander | Deployed the system, but if truly autonomous, did not control this individual act |
| The manufacturer | Designed hardware; typically shielded by commercial or sovereign immunity |
| The state | Possible under international law, but individual accountability disappears entirely |

(**hrw_losing_humanity_2012**)

# You Be the Judge: Who Is Responsible?

A loitering munition attacks a retreating fighter with **no human in the loop**. A civilian nearby is killed. An inquiry convenes. Every candidate has a defense.

**You have 100 points of responsibility. Distribute all of them:**

- **The machine** — but it has no agency to hold
- **The programmer** — who wrote the targeting logic two years and one war ago
- **The commander** — who launched it but did not choose this target
- **The manufacturer** — shielded by contract and by sovereign immunity
- **The state** — which can be liable while no individual ever is

**Then read your own answer back**

If every **human** on your list came out near zero, you have just described a machine that can kill lawfully and unaccountably. Is that a bug in your reasoning, or a fact about the weapon?

# Campaign to Stop Killer Robots

## The Campaign

- Founded in 2012 following the HRW report *Losing Humanity*
- Coalition of NGOs, academics, and former military officials
- Calls for a **preemptive ban** on fully autonomous weapons
- Analogous to bans on landmines (Ottawa Treaty, 1997) and cluster munitions (2008)

**Counter-argument**: States developing military AI (U.S., China, Russia) argue meaningful human control can coexist with autonomous speed and precision.


Campaign to Stop Killer Robots meeting, 2013. CC BY 2.0, via Wikimedia Commons.

# Case Study: The Kargu-2 in Libya (2020)

## The First Autonomous Lethal Attack?

In 2021, a UN Panel of Experts report described a 2020 incident in Libya:

- Government-allied forces deployed **Kargu-2** loitering munitions (Turkish-made drones with autonomous targeting)
- The drones were "programmed to attack targets without requiring data connectivity . . . no human was in-the-loop"
- This may have been the first recorded **lethal autonomous strike** on human targets

**Ambiguity**: Facts remain disputed; manufacturers denied full autonomy. But the threshold may have quietly already been crossed.

## Discussion

If fully autonomous weapons are already deployed, does the debate about whether to *allow* them still matter? What regulatory mechanisms remain available?

# Lecture Outline

# AI in the Lab: Opportunity and Dual-Use Risk

**AI has transformed biological research:**

- **AlphaFold** (DeepMind, 2021): predicted the 3D structure of nearly all known proteins, accelerating drug discovery and basic science
- AI-assisted genomics: faster pathogen identification and vaccine design
- AI-powered drug screening: reduces time and cost of finding therapeutic compounds

**But**: the same capabilities that speed up therapeutic research can speed up the design of harmful agents.

BSL-4 researcher in positive-pressure suit. Public domain (NIAID/NIH), via Wikimedia Commons.

# The Dual-Use Dilemma

**Dual-Use Research of Concern (DURC)**

Research that can both *protect* public health and *create* biological threats. The same knowledge is simultaneously beneficial and potentially catastrophic.

**sandbrink_ai_biological_misuse_2023** draws a critical distinction:

| **Type of AI** | **Dual-Use Risk Profile** |
|---|---|
| **LLMs** (GPT, Claude, etc.) | Primarily an *information hazard*: dangerous knowledge becomes more accessible, but little novel capability |
| **Biological design tools** (AlphaFold, RFdiffusion) | More serious: can design *novel* proteins, toxins, or pathogens that do not yet exist |

**Discussion**

Every capability on this slide has a legitimate use. Where would you actually **draw the line**—and who should be trusted to enforce it?

# Case Study: The Nerve Agent Experiment (2022)

## AI Designs VX Analogs in Hours

In 2022, researchers at Collaborations Pharmaceuticals published a disturbing result:

- They *inverted* their drug-discovery AI (designed to avoid toxic compounds) to instead maximize toxicity
- In under 6 hours, the model generated approximately **40,000 candidate molecules**
- Many were predicted to be **more toxic than VX nerve agent**
- The researchers halted the experiment and published a warning paper

**Key point**: the barrier to misuse is not primarily computational. It is biological and logistical — synthesizing and weaponizing such compounds still requires significant expertise and infrastructure.

(**sandbrink_ai_biological_misuse_2023**)

# Governance Responses to AI Biosecurity Risk

**Information Hazard**

An "infohazard" is information whose spread increases the probability of harm, even if the information itself is true and valuable in other contexts.

Current and proposed governance tools:

- **Sequence screening**: DNA synthesis companies screen orders for dangerous sequences (select agents, Australia Group pathogen list)
- **Model access controls**: restrict access to high-risk biological design AI; require biosecurity review for fine-tuning on pathogen data
- **Dual-use research oversight**: the U.S. NSABB reviews proposals liable to generate dangerous knowledge
- **AI guardrails**: major LLMs are trained to refuse dangerous biological requests — though jailbreaks and indirect elicitation remain possible

**The Core Tension**

The more powerful and broadly accessible AI tools become, the harder it is to prevent misuse without also suppressing legitimate research and scientific exchange.

# Lecture Outline

Part I: The Case for Existential Risk

Part II: The Critics Strike Back

Part III: Autonomous Weapons and AI in Warfare

Part IV: AI and Biological Threats

Part V: Longtermism and the Precautionary Principle

Part VI: Conclusion

# What Is Longtermism?

**Longtermism — MacAskill (2022)**

"We are living at the most important time in history . . . the decisions we make now could lock in outcomes for billions of years."

**Core claims**:

1. Future people matter morally as much as present people
2. Given low temporal discount rates, the long-run future contains *vastly* more people than the present
3. Therefore, improving the long-run trajectory of civilization is among our *most* important moral tasks
4. AI existential risk reduction is accordingly a top moral priority

**macaskill_future_2022**

The arithmetic: if humanity lasts one million years at current population, roughly $10^{14}$ future people exist. Even a 0.001% reduction in extinction probability "saves" millions of expected lives.

# The Expected Value Argument and Pascal's Mugging

## The Expected Value Case for X-Risk

- Let $P$ = probability of extinction from AI ($\sim$10% per Ord)
- Let $N$ = expected future humans if we survive ($\sim 10^{14}$)
- Expected lives at stake $= P \times N \approx 10^{13}$
- No near-term intervention rivals this in expected value
- **Conclusion**: AI safety deserves extraordinary moral priority

## Pascal's Mugging — The Skeptical Reply

- This logic can justify almost anything: multiply a tiny probability by a huge number and nothing else competes
- Probabilities of exotic future scenarios are not reliably estimable
- Expected value reasoning breaks down under radical uncertainty ("fanaticism" in decision theory)
- We have strong reasons to distrust intuitions about extremely large or small numbers

# The Precautionary Principle

**The Precautionary Principle (Wingspread Statement, 1998)**

When an action raises threats of harm to human health or the environment, **precautionary measures should be taken even if some cause-and-effect relationships are not fully established scientifically**. The burden of proof falls on those who claim the action is safe.

Applied to AI development:

- We do not need to *prove* that superintelligent AI will cause catastrophe
- We only need to establish that the risk is non-negligible and the potential harm enormous
- **Therefore**: slow AI development, mandate safety research, impose moratoria on dangerous capabilities

Precautionary logic underlies the Future of Life Institute's pause letter (2023), many EU AI Act provisions, and calls for a halt to frontier model training.

# Sunstein: The Precautionary Principle Is Incoherent

## Sunstein's Critique

The precautionary principle is paralytic because **every action *and* inaction carries risks**.

**Example**: Suppose we impose a moratorium on AI medical research because of dual-use risk:

- Cancer treatments that AI might have discovered are not developed
- Pandemic response tools are delayed
- These foregone benefits are *also* harms — harms of inaction

The precautionary principle tells us to be cautious about *both* paths, but cannot tell us *which* path to take. It provides no way to *compare* risks.

Sunstein's alternative: **cost-benefit analysis** — rigorously estimate and compare all costs and benefits of both action and inaction, including the costs of regulation itself.
(**sunstein_laws_of_fear_2005**)

# Comparing Approaches to AI Risk

| | Longtermist Focus | Present-Focused Justice |
|---|---|---|
| **Time horizon** | Centuries to millennia | Present to near-term |
| **Primary concern** | AI extinction risk | Bias, surveillance, labor exploitation |
| **Key thinkers** | Bostrom, Ord, Yudkowsky | Vallor, Gebru, Torres |
| **Main risk type** | Misaligned AGI, value lock-in | Power concentration; documented harms |
| **Policy priority** | Alignment research; moratoria | Regulation; account-ability; labor rights |

tension

## Discussion

Unfold the paper from the start of class. Did your number move? **What moved it**—an argument, or the room?

# Lecture Outline

## Synthesis: Both/And, Not Either/Or

We do not need to choose between taking existential risk seriously and addressing present-day harms. **But framing matters**:

- **"AI will kill everyone"** $\rightarrow$ resources flow to labs and frontier compute; little accountability for present bias and surveillance
- **"AI risk is just hype"** $\rightarrow$ present harms receive attention, but genuine long-term dangers may be neglected until it is too late
- **A better framing**: responsible AI development requires attending to risks at *multiple time horizons*, centering the most affected communities, and maintaining meaningful human control at every level

### The Political Economy Question

Who currently sets the agenda on AI risk? What voices are absent from the rooms where these decisions are made?

## Virtue Ethics and the Doomsday Debate

Vallor's technomoral virtues from Lecture 2 offer a compass for navigating genuine uncertainty:

| **Virtue** | **Applied to AI Risk** |
|---|---|
| **Humility** | Neither dismiss existential risk as science fiction nor claim certainty about timelines we cannot know |
| **Justice** | Ensure risk-reduction efforts do not themselves concentrate power or ignore present inequality |
| **Courage** | Raise hard questions about capability claims, safety theater, and who benefits from particular framings |
| **Care** | Center the needs of those most exposed to AI harms — in the present and in the future |

# Final Discussion: The $1 Billion Question

## Resource Allocation Under Uncertainty

You have $1 billion to allocate to reducing AI-related risks. How would you split it between:

(A) **AI alignment research**: technical work on ensuring advanced AI systems pursue human-beneficial goals
(B) **Autonomous weapons regulation**: international treaty development, export controls, accountability mechanisms
(C) **Present-day algorithmic bias**: auditing, litigation, regulation, and affected-community organizing around *existing* AI discrimination
(D) **AI biosecurity**: governance of biological design tools, sequence screening, dual-use oversight

**Justify your allocation.** What assumptions about probability, severity, and tractability drive your answer?

# Key Concepts i

**Existential risk** Any event that permanently annihilates Earth-originating intelligent life or drastically curtails its potential — the category Bostrom argues advanced AI may soon enter.

**Intelligence explosion** The hypothesis that an AI capable of improving its own design could rapidly bootstrap to vastly superhuman capability through recursive self-improvement.

**Orthogonality thesis** Intelligence and goal-content are independent: a superintelligent system could efficiently pursue *any* objective, including ones catastrophic for humans.

**Instrumental convergence** Whatever its final goal, a capable AI tends toward self-preservation, resource acquisition, and resistance to shutdown — without being explicitly programmed to.

**Alignment problem** The unsolved challenge of specifying human values precisely enough that a powerful optimizer pursues what we actually want rather than a damaging proxy.

**Stochastic parrot** Bender & Gebru's term for an LLM that produces statistically plausible text without understanding, semantics, or genuine intentions.

# Key Concepts ii

**TESCREAL** Gebru & Torres's acronym for the ideological bundle (Transhumanism, Extropianism, Singularitarianism, Cosmism, Rationalism, Effective Altruism, AI safety) whose assumptions are embedded in mainstream x-risk discourse.

**Longtermism** The view that shaping the long-run future is our most important moral task, because future people vastly outnumber people alive today. Championed by MacAskill; challenged by Torres.

**Pascal's Mugging** The objection that expected-value reasoning can justify almost any present sacrifice: multiply a tiny probability by an astronomically large future population and nothing in the present competes.

**Precautionary principle** Take precautionary measures against possible large harms even without full scientific proof, placing the burden on those claiming safety. Applied to AI; critiqued by Sunstein as incoherent.

**LAWS** Lethal Autonomous Weapons Systems — systems designed to select and engage targets without real-time human authorization.

# Key Concepts iii

**Meaningful human control** The ethical requirement that a human must be able to authorize each individual use of lethal force. The central contested standard in autonomous weapons governance.

**Accountability gap** The absence of any clear legal or moral responsible party when an autonomous weapon causes unlawful harm — machine, programmer, commander, manufacturer, and state all evade full responsibility.

**Dual-use research of concern (DURC)** Research that can simultaneously protect public health and enable biological threats; the same tool that designs vaccines can design toxins.

**Infohazard** Information whose spread increases the probability of harm regardless of its truth value — the key concept for evaluating whether LLMs pose biosecurity risks.

## The Core Tension

The existential risk debate is partly *empirical* (will AI become catastrophically dangerous?) and partly *distributional* (whose present concerns get displaced when we focus on speculative future harms?).

# References i